Sprint 2

In this Sprint, we iterated a bunch and did a lot of waiting around for our controllers to be built. Since I didn't have a controller to test with,I just tried out a bunch of sensors and did some sketches to see the most optimal way to control the steering of the horse. Another difficult input we are continuously changing is the jumping mechanic, which uses a force pressure sensor. While I was testing in the engine, I kept in mind how many times I jumped and how easy it was. There is simply too much going on the upper half of the body, and the focus on keeping the feet on the base of the horse while rocking proved to me too much to also throw in stomping the foot. To solve this, I want to try 1 more thing: putting the sensor on the seat and having the rider slightly lift their butt, but Plan B is to have auto jumping in the game, with the reason being having speed boosts that are triggered by jumping over hurdles similar to the speed rainbow platforms in Mario Kart.

Situation: Not enough work to be done this sprint while controllers were being built; continue to iterate on how a jump mechanic can be implemented while still feeling intuitive to use as a player.

Task: Continue researching different sensors and other ways we can use the jump mechanic. Figure out how to condense the hardware, so the final product is clean and wires are mostly out of sight.

Action: For jumping, I and the other engineers will test auto-jumping when you are in front of a hurdle and then apply a small speed boost. On the hardware side, I have two gyro sensors (one in front of the horse for steering and one in back for tracking rocking to convert to movement), but I am trying to eliminate the one in the back. I tested initially

with one in the front, but the accuracy wasn't there, so next sprint I will try again with the final controller when we bring it into the lab. The goal is to adjust the input sensitivity to have a good feel when rocking, not too hard, but enough to make you feel like rocking is rewarding you with going fast.

Result: As mentioned with testing, the accuracy was off, so with the new final controller I am hoping the result is much better, especially since the controller is better designed to accommodate the wiring. I have also looked into other gyro sensors for better accuracy in case there is too much inconsistency.

Last thing to note is the use of wifi boards. We talked with another team that is trying to go wireless, and the best thing to use is Bluetooth with a battery pack to keep the controllers/arduinos charged. This will be the most challenging step, but help from others and online resources will alleviate some of those problems.